Hafeez Khan

Email: mkhan@my.fit.edu | LinkedIn | Webpage | Google Scholar

Education

- **Florida Institute of Technology** — Florida, USA
  Master's & Ph.D. in Computer Science (PhD Advisor: Prof. Siddhartha Bhattacharyya) — *2023 - 2027 (Expected)*
- **Birla Institute of Technology and Science, Pilani** — Dubai, UAE
  Bachelors of Engineering in Computer Science (Thesis Advisor: Prof. Raja Muthalaghu) — *2019 - 2023*

Research Experience

Incoming Machine Learning Intern at RTX Technology Research Center (RTRC) — Fall 2026

- Topics: Agentic Large Languange Models, Post-training, RLHF, RLAIF, DPO, Formal Verification, OOD Detection

Systems Engineering Intern at Collins Aerospace — Summer 2026
Guide: Janet Liu, Cong Liu — *Collins Aerospace, USA*

- Topics: Path Planning, Representation Learning, Agentic AI, Trustworthy AI, Simulation, Explainable AI
- Core contributor to the **NASA ULI TRANSCEND** project; Developed a **representation learning-based** path planning model for runtime flight rerouting, improving planning efficiency by **86.67%** over the standard A* algorithm.
- Created an end-to-end simulation pipeline for flight rerouting and generated a synthetic dataset of over **100,000 flight scenarios with simulated airspace hazards.**
- Developed explainable AI methods to assess model robustness and provide reliable rerouting recommendations for pilots.

Research Assistant at ASSIST Lab (Florida Tech) — Fall 2023 – Present

- **NASA-Funded Research** — Fall 2024 – Present
  Guide: Natasha Neogi, Sarah Lehman, Siddhartha Bhattacharyya, Philip Chan — *NASA LaRC & Florida Tech*
  - Topics: World Models, Self-Supervised Learning, Multi-modal Learning, Diffusion Models, Reinforcement Learning
  - Developed a self-supervised world model, improving **ImageNet-1k Top-1 accuracy by 2.1%** over baselines with linear probing, and achieving similar gains on semantic segmentation, object and OOD detection benchmarks.
  - Created a post-training pipeline for the flight path-planning agent using DPO, reducing safety constraint violations by **12%** while maintaining **91%** planning efficiency.
  - Developed a test-time input refinement framework for few-shot segmentation, improving mIoU by up to **36%** across five diverse benchmark datasets, including Pascal-5i and COCO-20i.
- **Microsoft Research (Collaboration)** — Spring 2025
  Guide: Yash Jain, Vibhav Vineet — *Microsoft Research, USA*
  - Topics: Image Generation, Multi-modal Large Language Models, Diffusion Models, Text-to-Image, Optimization
  - Developed a closed-loop test-time prompt refinement framework using multimodal LLMs, improving text-to-image alignment by **11.7%** on DALL·E 3, **12.9%** on FLUX, and up to **15.25%** on Stable Diffusion 1.5/2.1/3 models.
- **DARPA-Funded Research** — Summer 2025
  Guide: Junaid Babar, Isaac Amundson, Siddhartha Bhattacharyya — *Collins Aerospace, USA*
  - Topics: Trust and Safety, Model Checking, Formal Verification, Reinforcement Learning, Language Translation
  - Developed an automated Soar-to-PRISM translator using ANTLR and Java, enabling probabilistic model checking and formal verification of cognitive agents while exposing safety violations undetectable through testing alone.
- **American Heart Association-Funded Research** — Summer 2024
  Guide: Venkat Keshav Chivukula — *UTHealth Houston, USA*
  - Topics: 3D/4D Reconstruction, Representation Learning, Medical Imaging, Computational Fluid Dynamics (CFD)
  - Developed a 3D autoencoder for intraventricular velocity reconstruction, outperforming UNet3D by **10.55%** PSNR.
  - Generated synthetic CFD datasets across **eight LVAD patient geometries** to train 3D models.

Awards

- **Awarded 1st Place by Meta** in the Computationally Optimal Gaussian Splatting Competition at the IEEE/CVF International Conference on Computer Vision (ICCV) 2025. 2025
- Awarded for **Outstanding Academic Achievement** at Florida Tech. 2025
- **Undergraduate Research Award** for outstanding Bachelors Thesis at BITS Pilani Dubai. 2023
- **Gold Medalist** in IEEE Robotics Competition, 3D Object Detection and Tracking for Road Safety, held in Dubai. 2023

Skills

Programming Languages and Web Apps: Python, R, Matlab, C/C++, Java, SQL, Flask, Streamlit

Proficient Deep Learning and Vision Libraries: PyTorch, TensorFlow, HuggingFace, Transformers, OpenCV

Cloud Platforms & Services: Azure (ML Studio, DevOps), AWS (S3, SageMaker), GCP (Google Maps, Cloud Storage).

Selected Publications

∗ equal contribution

It Is Not the Model, It Is the Input: Test-Time Input-Space Refinement for Few-Shot Segmentation
M.A. Hafeez Khan, Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
Under Review at an A* Conference, 2026

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift [PDF]
M.A. Hafeez Khan, Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
[**Oral Presentation**] International Conference on Computer Vision 2COOOL Workshop (**ICCV**), 2025

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding in Diverse Conditions [PDF]
Parth Ganeriwala, M.A. Hafeez Khan, Amy Alvarez, Siddhartha Bhattacharyya, Natasha Neogi, Sarah M. Lehman
American Institute of Aeronautics and Astronautics (**AIAA**) SCITECH Forum, 2026

Test-time Prompt Refinement for Text-to-Image Models [PDF]
M.A. Hafeez Khan*, Yash Jain*, Siddhartha Bhattacharyya, Vibhav Vineet
[**Invited Talk**] International Conference on Computer Vision MARS2 Workshop (**ICCV**), 2025

Few-Shot Classification and Anatomical Localization of Tissues in SPECT Imaging [PDF]
M.A. Hafeez Khan, Samuel M. Boddepalli, Siddhartha Bhattacharyya, Debasis Mitra
[**Oral Presentation**] Medical Imaging Conference (**MIC**), 2025

LVADNet3D: A Deep Autoencoder for Reconstructing 3D Intraventricular Flow from Sparse Data [PDF]
M.A. Hafeez Khan, Marcello Mattei, Ben Diaz, Ruth White, Siddhartha Bhattacharyya, Venkat Keshav Chivukula
[**Oral Presentation**] International Conference on Machine Learning and Applications (**ICMLA**), 2025

NORA: A Nephrology-Oriented Representation Learning Approach Towards CKD Classification [PDF]
M.A. Hafeez Khan, T. Bhattacharyya, O. Khan, Noorah Khan, Alina Khan, M.Q. Khan, Sujoy Hajra
[**Spotlight Paper**] International Conference on Machine Learning and Applications (**ICMLA**), 2025

ALINA: Advanced Line Identification and Notation Algorithm [PDF]
M.A. Hafeez Khan, Parth Ganeriwala, Siddhartha Bhattacharyya, Natasha Neogi, Raja Muthalagu
IEEE / CVF Computer Vision and Pattern Recognition Conference (**CVPR**), 2024

A Hybrid BiLSTM-CNN Approach for Intrusion Detection for IoT Applications [PDF]
M.A. Hafeez Khan, Sapna Sadhwani, Raja Muthalagu, Pranav Pawar, K. Suresh
Scientific Reports, Springer, 2024

Classification of Microstructure Images of Metals Using Transfer Learning [PDF]
M.A. Hafeez Khan, Hrishikesh Sabnis, J. Angel Arul Jothi, J. Kanishkha, A.D. Prasad
International Conference on Modelling and Development of Intelligent Systems (**MDIS**), 2022

Detection of Cavities from Oral Images using Convolutional Neural Networks [PDF]
M.A. Hafeez Khan, Giri Prasad S., J. Angel Arul Jothi
[**Best Paper Award**] IEEE International Conference on Electrical, Computer and Energy Technologies (**ICECET**), 2022